

Experiment 5: Perceptron vs Multilayer Perceptron

B.E CSE - Machine Learning Lab

Aim

To implement and compare the performance of Perceptron Learning Algorithm (PLA) and Multilayer Perceptron (MLP) with hyperparameter tuning.

Objective

- Implement PLA and MLP
- Perform hyperparameter tuning
- Compare performance using evaluation metrics

Dataset

English Handwritten Characters Dataset:

- Total samples: 3410
- Classes: 62 (0-9, A-Z, a-z)
- Type: Grayscale images

Theory

Perceptron Learning Algorithm

$$w_{t+1} = w_t + \eta(y - \hat{y})x \quad (1)$$

- Linear classifier
- Uses step activation
- Works only for linearly separable data

Multilayer Perceptron

- Uses hidden layers
- Activation: ReLU, Sigmoid, Tanh
- Optimizers: SGD, Adam
- Learns nonlinear patterns

Implementation

PLA

- Implemented from scratch
- Used step activation
- Applied One-vs-Rest for multi-class

MLP

- Hidden layers used
- Activation: ReLU
- Optimizer: Adam

Hyperparameter Tuning

Layers	Activation	Optimizer	LR	Batch	Accuracy
1 (128)	ReLU	SGD	0.01	32	0.94
2 (256-128)	ReLU	Adam	0.001	64	0.97
3 (512-256-128)	Tanh	Adam	0.0005	64	0.96

Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
PLA	79.44	0.79	0.79	0.79
MLP	96.94	0.97	0.97	0.97

Results

- MLP performed better than PLA
- PLA failed to learn nonlinear patterns
- MLP showed stable convergence

Observations

- PLA underperforms due to linear limitation
- Learning rate and optimizer had major impact
- More layers improved accuracy but risked overfitting

Conclusion

MLP significantly outperforms PLA in handwritten character recognition due to its ability to learn nonlinear decision boundaries. Hyperparameter tuning improves performance and convergence.